



The symmetric technique of formant transition generation for use in speech synthesis in Arabic

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Received: 1 March 2024 / Accepted: 31 May 2024 / Published online: 24 July 2024
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Abstract The objective of this work is to elaborate on a vocal communication aid system for blind individuals in the Arab world, which is also used as a system for learning Arabic for both individuals and nonnative speakers. This system is a result of the development of a speech synthesizer for the Arabic language, which is based on the concatenative synthesis method (end-to-end speech data collection (Eskenazi et al. <https://ieeexplore.ieee.org/servlet/opac?bknum=8039813>, John Wiley & Sons, 2013) of subsyllable sound units typically stored in digital wave format (Vaseghi in Advanced digital signal processing and noise reduction, John Wiley & Sons, Hoboken, 2008), obtained by segmentation using software called PRAAT (Gold et al. in Speech and audio signal processing: processing and perception of speech and music, John Wiley & Sons, Hoboken, 2011; Pleva et al. in 2015 25th international conference radioelektronika (radioelektronika), pp 172–175, <https://doi.org/10.1109/RADIOELEK.2015.7129000>) from a natural language corpus (Chou et al. in IEEE Trans Speech Audio Process 10:481–494, 2002). The treatments in this study took place in different phases (in MUDALLA: PROCEEDING INTERNATIONAL CONFERENCE ON ARABIC LANGUAGE, 1(1): 1590–1595, 2020 <http://arab.sastra.um.ac.id/wp-content/uploads/2021/10/PROCEEDING-INTERNATIONAL-CONFERENCE-ON-ARABIC-LANGUAGE-FULL.pdf>). Text analysis techniques are responsible for converting incoming text into a linguistic representation that encodes information about how the input text should be pronounced (Shiga et al. in Text-to-speech synthesis, pp 39–52, 2020 https://doi.org/10.1007/978-981-15-0595-9_3). The method used to produce missing phonetic units in syllable counts [CVs: consonant–vowel] is to use mathematical symmetry from [VCs: vowels–consonant], which results in the construction of a phonetic lexicon based on a reduced number of phonemes, which is 124, the resulting patterns are enhanced as transition envelopes of [VCs] generated from their counterparts [CVs] using this feature. Finally, after including the linear smoothing stage (Dutoit in IEEE international conference on acoustics, speech and signal processing, 1: I/565–I/568, 1994 <https://doi.org/10.1109/ICASSP.1994.389231>), speech outputs that resemble natural speech as much as possible without noise are obtained (Millstein in Natural language processing with python: natural language processing using NLTK <https://realpython.com/nltk-nlp-python/>, Frank Millstein, 2020); these outputs can be continuous and intelligible at different levels of language that can be employed in synthesizing intelligible speech for blind individuals in the Arab community, noting that this method is mathematically useful for all languages of the world in terms of producing the quality of speech.

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Keywords Text-to-speech · Arabic units · Speech synthesis · Concatenative synthesis · Generated forms · Symmetric technique

1 Introduction

Arabic ranks fourth among all languages used worldwide [10]. It also has religious value worldwide for more than 1.6 billion Muslims [11], with nearly 500 million speakers spread across 25 countries, and given this global status, many researchers are interested in studying, and learning, and teaching this language. Two groups of researchers seek to integrate it into the world of technology, a group that seeks to expand the scope of marketing their various products to the largest possible number in the world to achieve their commercial goals, especially after Arabic was adopted among the working languages of the United Nations [12], and a group that seeks to remove backwardness and keep pace with what the rest of the countries of the world are witnessing, which is experiencing a qualitative leap. Their surroundings are smart with various applications to achieve their development goals.

Due to the diversity of applications, such as: artificial intelligence, machine learning, deep learning, speech to machine and many other applications [13, 14], the speech synthesizer, also known as the text-to-speech (TTS) system [15], is one of the most important current technologies that enables us to read emails, text messages, and other human–machine interactions using multimedia applications. People who are deaf and have a voice impairment can use them as primary reading devices to interact with others who do not understand sign language. It is useful and popular among visually impaired people. It can also be applied for a variety of educational purposes, including teaching foreign language pronunciation and using speech therapy.

How speech is synthesized using different techniques and in different languages has been widely studied. A neural network that learned to pronounce English text correctly was created in 1987 by Sejnowsky and Rosenberg. Strings of English text are converted into strings of phonemes by the system known as NETtalk, which is built using a large number of parallel network systems capable of capturing a large number of regularities and many irregularities in English pronunciation [16]. Two neural networks were used to create a rule-based system by Karaali et al. [17], which translates the phonetic representation of speech into an audio representation and then into speech. The first method uses a time-delay neural network. The second is used to regulate the timing of speech output.

The top 5 TTS software programs available today include Balabolka [18, 19], NaturalReader [20], Lecteur de Réves Vocal [20], Amazon Polly [21], and Speechify [22]. Regarding to the Arabic language, there are as of now a few commercially accessible TTS frameworks such as ARABTALK, BrightSpeech, ElanSpeech and Sakhr TTS accessible from

Research and Development International (RDI), Babel Advances and Babel-Infovox as stated in [23], and an important topic in the field of Natural Language Processing (NLP) for generating chat with robot in Arabic [24].

Among the text-to-speech applications in the field of communications services, text-to-speech applications are important in relation to services provided to people special needs [25]. For example, developing a voice communication support system for a patient living in an Algerian environment could include encoding, compression and synthesis systems, particularly those implemented through so-called TTS [6], and there is personal control systems such as the one developed by Salah Abdelrahman et al. [26], it works using Arabic voice commands for the elderly and people with disabilities.

Deep learning has revolutionized artificial voice production systems. Two key elements, in particular, have seen significant innovation: the process of converting text into high-level phonetic representations and the process of converting those representations into speech. Two deep learning paradigms, WaveNet in 2016 [27] and Tacotron in 2017 [28, 29], have provided significant improvements over existing systems because deep learning is now fundamental to modern speech platforms.

Sequential speech synthesis systems mainly involve algorithms that associate selected units and perform signal processing, such as smoothing sequence boundaries. They also involve selecting appropriate units from the speech repertoire [30]. The purpose of this method is to convert texts into audio sequences through linear smoothing in each syllable from first sequence to a higher sequence to increase the quality of synthesis speech of the Arabic language, according to an algorithm based on the synthesis of audio units of the type (Consonant half-Vowel) generated from the reference syllable [CVCC], for a small number of units present in the phonetic lexicon with the possibility of generating reduced units with a mathematical symmetry, while ensuring the presence of formant transitions responsible for the continuity and quality of the speech produced.

2 Methods

In this section, we will conduct a study on the possibility of using an existing syllable in the list of syllables obtained from segmenting the studied corpus to generate a new syllable that does not already exist in the list, provided that formant transitions are preserved for the generated syllable. Symmetry is a generative property of models from their originals that depends on a particular reference. In this study, all the [VC] type syllables are used to produce all the

[CV] type units, and vice versa. This approach performs the same function as a mirror, which casts an image that is completely the opposite of the source. Arabic contains several (28) consonants, 7 consonants that are always accentuated consonants AACs, 19 consonants of them are always diluted consonants ADCs, and 2 consonants are sometimes diluted, and sometimes are accentuated called consonants depending on the context CDCs [31].

In this study, we use only $28 \times 3 = 84$ subsyllables in total, and they are of the type Consonant half-Vowel [CV_h]. However, all the rest are suffixes, which are defined as follows:

- We use six open half-vowels, three of which are normal and three of which are emphasized [V_{ho(n+e)}].
- We use six closed half-vowels, three of which are normal and three of which are emphasized [V_{hc(n+e)}].
- The last consonant in [CVCC] syllables can be used for suffixes, although postconsonants are usually longer than consonants. There are 28 of these consonants [32].
- The composition of full open vowels, which are 6 vowels (three normal and three pharyngealized vowels), is the result of doubling the previously mentioned half vowels, which are used to produce syllables that have long vowels (CVVs).
- This strategy employs 84 (28×3) units of type [CV], which, in the cases of Fat'ha (a), Dhamma (u), and Kasra (i), are concatenated with 28 phonemes.

2.1 Application of the method of schemes to the symmetric technique of generation

The following algorithm summarizes the method used in this research so that the following diagram contains all the stages necessary for speech synthesis. This method is part of the development of the TTS system for the Arabic language. The phonetic spelling transcription phase is considered a starting point for the process of automatic speech synthesis using this technique. This is done through several phases, as illustrated in the diagram, which shows all the cases necessary for this and how to treat them according to a specific algorithmic path (Fig. 1).

The representative schemes of the forms are very generative of the morphological units of the Arabic language. The purpose of the technique involving symmetry (envelopes carrying formantic transitions) represented by generative schemes is to develop a system that can convert written messages into audio messages. This system uses these schemes as an intermediate phase or a passage between the phonetic orthographic (spelling) transcription and the concatenation of the corresponding sounds of the written sequences received. This passage must guarantee text-to-speech

conversion for all possible cases of the Arabic language, gemination, emphasis, and long and brief vowels in the declarative state of the sentence without taking into consideration the characteristics of the intonation, which is akin to exclamatory, interrogative and negative sentences. The generative scheme or the representative form is based on the form of the reference syllable [CVCC], which is represent the general form to generate all the syllables in the Arabic language according to Eq. (1) in the following manner:

$$C_n V_n C_{nn} C_{nnn} = CVCC \begin{bmatrix} C_n \\ 1 \\ C_{nn} \\ C_{nnn} \end{bmatrix} \quad (1)$$

All the syllable types are generated by the scheme of the reference syllable [CVCC] using Eq. (1), and the value vectors of the consonants contain $[C_n, C_{nn}, C_{nnn}]^T$. Two types of expressions for consonants are illustrated in Table 1:

These characteristics can be transcribed via the symmetric method. Taking the following example to understand the stage of generation step by step, the previous sequence [yabḏulna juhdaḥunna بيذّلن جدهن] and the sound part are obtained as follows:

$$S_{sp}(C) = \sum(yabḏulna juhdaḥunna) = \sum([yab],[ḏul],[na],[juh],[da],[hun]).$$

We introduce the representation by the reference syllable, and we obtain Eq. (2):

$$S_{sp}(C) = \sum \left\{ \left(C[a]CC \begin{bmatrix} y \\ 1 \\ b \\ 0 \end{bmatrix}, C[u]CC \begin{bmatrix} ḏ \\ 1 \\ 1 \\ 0 \end{bmatrix}, C[a]CC \begin{bmatrix} n \\ 1 \\ a \\ 0 \end{bmatrix} \right) + \left(C[u]CC \begin{bmatrix} j \\ 1 \\ h \\ 0 \end{bmatrix}, C[a]CC \begin{bmatrix} d \\ 1 \\ 0 \\ 0 \end{bmatrix}, C[u]CC \begin{bmatrix} h \\ 1 \\ n \\ 0 \end{bmatrix} \right) \right\} \quad (2)$$

We introduce the representation of dependence on the left and right sides; we obtain Eq. (3):

$$S_{sp}(C) = \sum \left\{ \left(\left(C[a]CC \begin{bmatrix} y \\ 1 \\ 0 \\ 0 \end{bmatrix} + C[a]CC \begin{bmatrix} 0 \\ 1 \\ b \\ 0 \end{bmatrix} \right) \left(C[u]CC \begin{bmatrix} ḏ \\ 1 \\ 0 \\ 0 \end{bmatrix} + C[u]CC \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} \right) C[a]CC \begin{bmatrix} n \\ 1 \\ 0 \\ 0 \end{bmatrix} \right) + \left(\left(C[u]CC \begin{bmatrix} j \\ 1 \\ 0 \\ 0 \end{bmatrix} + C[u]CC \begin{bmatrix} 0 \\ 1 \\ h \\ 0 \end{bmatrix} \right) \left(C[a]CC \begin{bmatrix} d \\ 1 \\ 0 \\ 0 \end{bmatrix} + C[a]CC \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \right) C[u]CC \begin{bmatrix} h \\ 1 \\ n \\ 0 \end{bmatrix} \right) \right\} \quad (3)$$

We apply the symmetric technique, and obtain Eq. (4):

Fig. 1 Algorithm for speech synthesis via the symmetric method

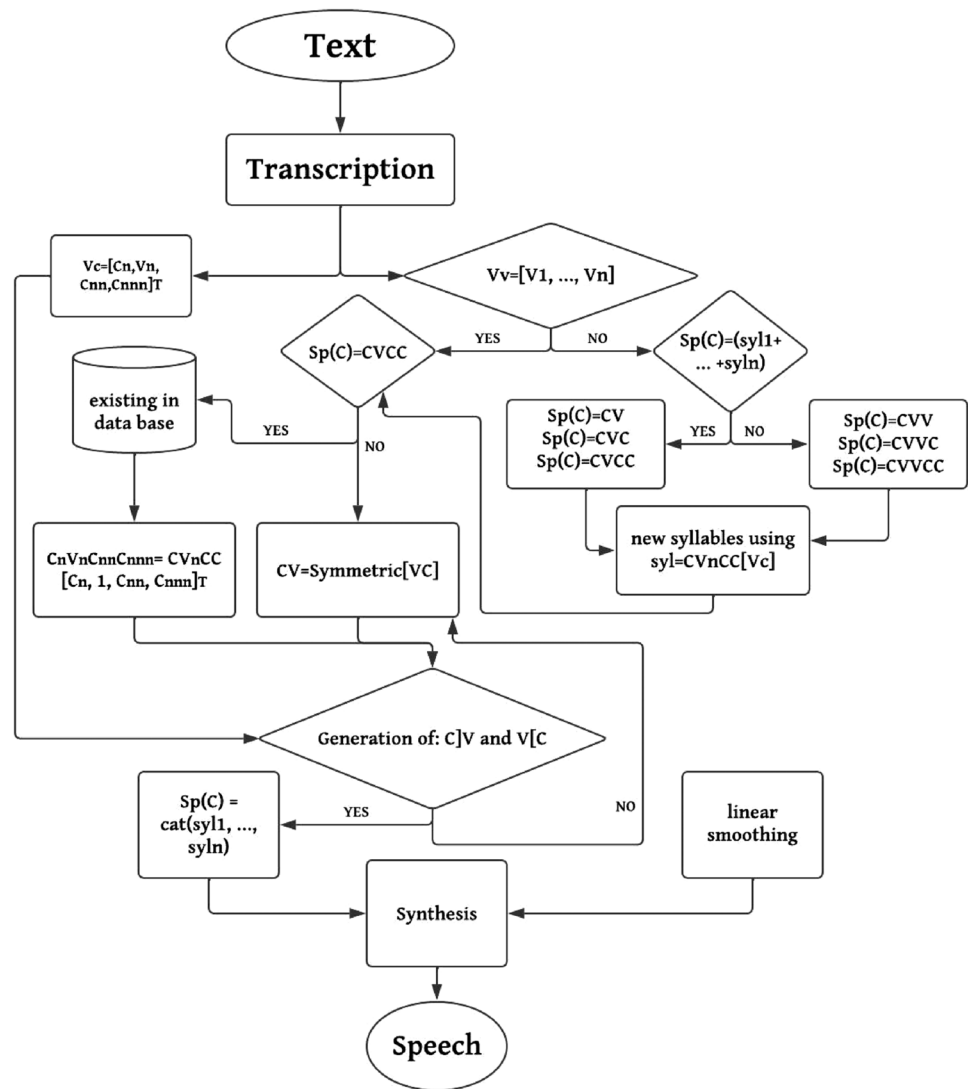


Table 1 Generation of new syllabic forms by the reference syllable $[CV_nCC]$

Brief vowel syllables	Long vowel syllables
$C_1V_1 = C_1CC \begin{bmatrix} C_1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$	$C_1V_1V_2 = CV_nCC \begin{bmatrix} C_1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + CV_2CC \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$
$C_1V_1C_{1,1} = C_1V_1CC \begin{bmatrix} C_1 \\ 1 \\ C_{1,1} \\ 0 \end{bmatrix}$	$C_1V_1V_2C_{2,2} = CV_1CC \begin{bmatrix} C_n \\ 1 \\ 0 \\ 0 \end{bmatrix} + CV_2CC \begin{bmatrix} 0 \\ 1 \\ C_{2,2} \\ 0 \end{bmatrix}$
$C_1V_1C_{1,1}C_{1,1,1} = CVCC \begin{bmatrix} C_1 \\ 1 \\ C_{1,1} \\ C_{1,1,1} \end{bmatrix}$	$C_1V_1V_2C_{2,2}C_{2,2,2} = CV_1CC \begin{bmatrix} C_1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + CV_2CC \begin{bmatrix} 0 \\ 1 \\ C_{2,2} \\ C_{2,2,2} \end{bmatrix}$

$$S_{sp}(C) = \sum \left\{ \left(\left[Ca[CC \begin{bmatrix} y \\ 1 \\ 0 \\ 0 \end{bmatrix} + \tilde{Ca}CC \begin{bmatrix} 0 \\ 1 \\ b \\ 0 \end{bmatrix} \right], \left[Cu[CC \begin{bmatrix} \partial \\ 1 \\ 0 \\ 0 \end{bmatrix} + \tilde{Cu}CC \begin{bmatrix} 0 \\ 1 \\ l \\ 0 \end{bmatrix} \right], C[a[CC \begin{bmatrix} n \\ 1 \\ 0 \\ 0 \end{bmatrix}] \right) \right. \\ \left. + \left(\left[Cu[CC \begin{bmatrix} j \\ 1 \\ 0 \\ 0 \end{bmatrix} + \tilde{Cu}CC \begin{bmatrix} 0 \\ 1 \\ h \\ 0 \end{bmatrix} \right], C[a[CC \begin{bmatrix} d \\ 1 \\ 0 \\ 0 \end{bmatrix}], \left[Cu[CC \begin{bmatrix} h \\ 1 \\ 0 \\ 0 \end{bmatrix} + \tilde{Ca}CC \begin{bmatrix} 0 \\ 1 \\ n \\ 0 \end{bmatrix}] \right) \right) \right\} \quad (4)$$

All the syllables are of the form $(C_n V_n C_{nn} C_{nnn})$, and all values corresponding to (C_{nn}) and (C_{nnn}) are null; we can reduce them from these vectors to get Eq. (5):

Conversely, in Eq. (6) we can generate this sequence using consonants values corresponding:

$$S_{sp}(C) = \sum \left\{ \left(\left[Ca \left[\begin{bmatrix} y \\ 1/2 \end{bmatrix} \right], \left[\tilde{Ca} \left[\begin{bmatrix} b \\ 1/2 \end{bmatrix} \right], \left[\tilde{Cu} \left[\begin{bmatrix} \partial \\ 1/2 \end{bmatrix} \right], \left[\tilde{Cu} \left[\begin{bmatrix} 1 \\ 1/2 \end{bmatrix} \right], C[a \left[\begin{bmatrix} n \\ 1 \end{bmatrix} \right] \right) \right. \right. \\ \left. \left. + \left(\left[Cu \left[\begin{bmatrix} j \\ 1/2 \end{bmatrix} \right], \left[\tilde{Cu} \left[\begin{bmatrix} h \\ 1/2 \end{bmatrix} \right], C[a \left[\begin{bmatrix} d \\ 1 \end{bmatrix} \right], \left[Cu \left[\begin{bmatrix} h \\ 1/2 \end{bmatrix} \right], \left[\tilde{Cu} \left[\begin{bmatrix} n \\ 1/2 \end{bmatrix} \right] \right] \right) \right) \right\} \quad (5)$$

It is a representation according to syllables $[CV]$ and symmetric syllables $[CV]$, which are of the same type.

$$S_{sp}(C) = \sum \left(\left(ya \left[\begin{bmatrix} \cdot \\ \cdot \end{bmatrix}, \tilde{ba} \left[\begin{bmatrix} \cdot \\ \cdot \end{bmatrix} + \partial u \left[\begin{bmatrix} \cdot \\ \cdot \end{bmatrix} \right] \right] \right) \right. \\ \left. + \left(ju \left[\begin{bmatrix} \cdot \\ \cdot \end{bmatrix}, \tilde{hu} \left[\begin{bmatrix} \cdot \\ \cdot \end{bmatrix} \right], da \right] + hu \left[\begin{bmatrix} \cdot \\ \cdot \end{bmatrix} \right] \right) \right) \quad (6)$$

By using the results of the symmetric method in Eq. (6), we have Eq. (7):

$$S_{sp}(C) = \sum \{ (ya[\cdot], [ab] + (\partial u[\cdot], [ul] + na)) + (ju[\cdot], [uh] + da) + (hu[\cdot], [un]) \} \\ = \sum \{ (y)a[b] + (\partial)u[l] + na) + (j)u[h] + da + (h)u[n] \} \quad (7)$$

As a result of generation, we have Eq. (8):

$$S_{sp}(C) = \sum \{ (yab), (\partial ul), na + (juh), da, (hun) \} \quad (8)$$

Table 2 Symmetry generation test sentences

No	Sentences	No	Sentences
1.	TafaHa-l-kayl	2.	Lam-yakun-carisan
3.	Eavhabu-bi-Eamaanin	4.	BaOaVta-fiina-naviiran
5.	LavaOtahu-biqawlin	6.	SubHaanaka
7.	Kuntu-qudwatan-lahum	8.	Balaga-s-saylu-z-zubaa
9.	Lam-yastamtiO-bi-Vamarihaa	10.	Li-v-vakari-miVlu-Happi-l-EunVayayni
11.	kaana-fii-pulumaatin-wa-lam-YarHal	12.	Yabðulna juhдахunna
13.	Yuqaaamiruuna-bil-maal	14.	Liyajðibna-l-Habl
15.	Kaana-SaaEiman	16.	faabbaan-ʔiθnaan
17.	Fastagfara-li-vanbihi	18.	Li-S-Saaʔimi-farHataan
19.	Eaxava-Eijazatan	20.	YartaTimu-bil-jidaar

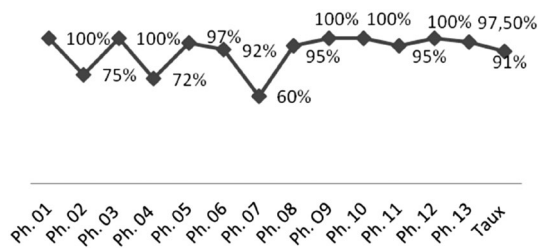
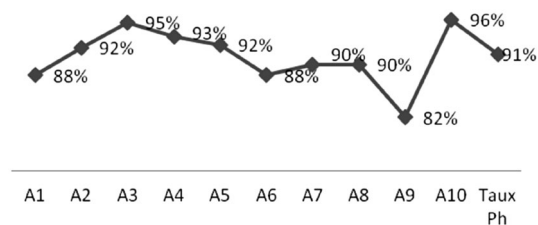
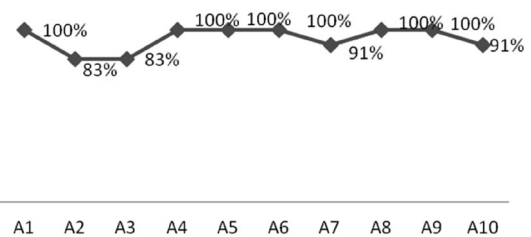
3 Results and discussion

We prepared a set of sentences to evaluate the results obtained from this technique, and we proposed them for ten people to hear because the numerical results are very reliable when choosing the technique or when commenting on it. The following sentences are suggested for evaluating the efficacy of this technique, as illustrated in Table 2.

Table 2 contains 20 sentences in the Arabic language, 12 sentences were selected from [33], and 8 were proposed

Table 3 Hearing results for each listener and for each sentence

Sentence	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	Rate 1%
1	1	1	1	1	1	1	1	1	1	1	100
2	0.5	1	1	1	0.5	0.5	0.5	0.5	1	1	75%
3	1	1	1	1	1	1	1	1	1	1	100
4	0.65	0.65	0.65	0.65	1	0.65	0.65	0.65	1	0.65	72
5	1	1	1	1	1	1	1	1	1	0.7	97
6	0.8	0.8	1	1	1	0.8	1	1	0.8	1	92
7	0.5	0.5	0.5	0.5	0.5	0.5	1	0.5	1	0.5	60
8	1	1	1	1	1	1	1	1	0.75	1	98
9	1	1	1	1	1	0.75	1	1	1	1	98
10	1	1	1	1	1	1	1	1	1	1	100
11	1	1	1	1	1	1	1	1	1	1	100
12	1	1	1	1	1	1	1	1	1	1	10
13	1	1	0.75	1	1	1	1	1	0.75	1	95
14	1	1	1	1	1	1	1	1	1	1	100
15	1	1	1	1	1	1	1	1	0.75	1	98
16	1	1	1	1	1	1	1	1	1	1	100
17	0.8	1	1	1	1	0.93	0.8	0.93	1	1	95
18	1	1	1	1	1	1	1	1	1	1	100
19	1	1	1	1	1	1	1	1	1	1	100
20	1	1	1	1	0.88	1	1	0.88	1	1	98
Rate 2	91%	94%	94%	96%	94%	90%	94%	92%	95%	94%	93

**Fig. 2** Recognition rate for each sentence**Fig. 3** Recognition rate for each listener**Fig. 4** Hearing test results for [yuqaamiruuna] after correction

gave the hearing results inserted in Table 2, with the symbolization of listeners from L1 to L10.

Table 3: represents the recognition rate for each listener's of the sound sentences synthesized using the symmetric method, on the one hand, and the rate of all group members' recognition of the same words in the speech synthesis of sentences, on the second hand, and the rate of the whole group's recognition of the synthesized speech of words for all of the selected sentences, on the other hand, which is a general value for these results. The results are included in Fig. 2, which shows the identification rate of sentences that have been synthesized as a curve.

According to Fig. 2: recognition rate for each sentence, the sound produced for sentences 4 and 7 was recognized at a rate of less than 75% compared to the other sentences, except for the second sentence, which had a rate of 75% compared to the content of the written sequence of the

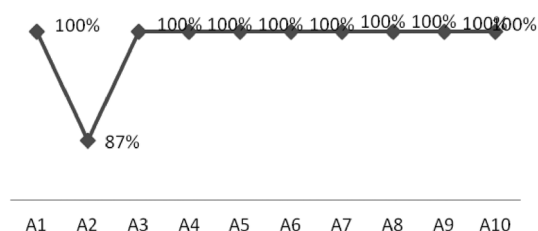


Fig. 5 Hearing test results for [qudwatan] after correction

sentence. This rate represent misidentification of the sound corresponding to the sentence occurred due to the use of a syllable of the inverse type [ha] from the bad segmentation with a sound nature close to [Ha], which generates a bad sound for the word [ʔaḏhabu] and is heard as [ʔaḏHabu]. The correction performed finds a new syllable that corresponds to [ha], which is sufficient to hear the resulting sound corresponding to the specified sequence. In the next curve, we have the identification rate for all sentences for each listener. The lowest score in this case is 82% for listener nine (see Fig. 3: recognition rate for each listener), an average third-year student. Because the rate is very low and because it does not accurately reflect the generated result, we must make corrections so that each listener can accurately recognize the tuning content.

3.1 Correction of misidentified sequences

The inaccuracy in recognizing the sequence of words [qudwatan] and [yuqaamiruuna] is caused by the presence of poor-quality phonetic units [wa] and [ruu], which do not fit into this sequence (inappropriate place) because the sound [w] is emphasized, which is similar to the French [oi] sound and creates an emphasis that extends throughout the entire sequence, making it difficult to perceive the true sound of the string unit others mistakenly hear [n] for [d] and so on for the outcomes.

We searched for the extraction of a sound unit of type [wa], found a unit of type [kaw], applied symmetry to this unit to obtain the opposite sound, which is [wak], and deleted the sound of [k] from the syllable to have [wa] at the end of this segmentation, to make a good correction of the result of concatenation and to obtain a good generation that serves to give sound units corresponding to the written sequences.

The same procedure was used to obtain a syllable unit [ru], which is also very helpful in producing the sound of the word [yuqaamiruun], which is significantly more palatable to listeners than the previous time.

After the correction, for the same sequence in the seventh sentence, the identification rate of the generated sequences

Table 4 Comparison between polyphem method speech synthesis and symmetric method speech synthesis [33]

Sentences	Polyphem method speech synthesis			Symmetric method speech synthesis	
	Test 1	Test 2	Test 3	Concat-enation using [CV _n]	Regenerate by symmetric method
1	20.0	37.5	37.5	100	100
2	67.5	81.3	87.5	75	87.5
3	25	31.3	42.5	100	100
4	100	100	100	72	100
5	93.8	97.5	100	97	97
6	100	100	100	92	92
7	97.5	98.3	98.3	60	98
8	99.6	100	100	98	98
9	95.8	95.8	95.8	98	98
10	100	100	100	100	100
11	100	100	100	100	100
12	83.8	90	100	100	100
13	100	100	100	95	95
14 with 9 in [33]	76.3	93.8	95	100	100
15 with 19 in [33]	90.8	92.5	95.8	98	100
16 with 20 in [33]	75.8	88.3	91.7	100	100
17 with 5 in [33]	96.3	97.5	97.5	95	95
18 with 11 in [33]	51.3	73.1	85.6	100	100
19 with 8 in [33]	58.8	84.4	93.8	100	100
20 with 2 in [33]	93.8	98.8	98.8	98	98
Rate	81.2	88	90.9	94	97.9

of [yuqaamiruuna] increased to more than 83% for the second and third listeners, to 91% for the seventh and tenth listeners, and to 100% for all the other listeners. The group's listener recognition rate is 94.8%, which is an extremely positive outcome for this kind of Arabic speech synthesizer (see Fig. 4).

After the correction, the second listener's rate of correctly identifying the sequence [qudwatan] in the fourth sentence increased to more than 87%, and the rate for all the other listeners was 100%. Thus, the typical rating for this group is 98.7%, as indicated in Fig. 5.

The results in Fig. 5: show this method of synthesis gave good results, and they also show that correction sound units by symmetry technique produced effective results.

3.2 Comparison between the symmetric technique and the polyphem arabic speech synthesis method

In this section, we compare the results of Arabic speech synthesis using the symmetric technique to generate $[CV_h]$ with the results of linear smoothing of a hybrid composition system by sequencing polyphems [33]. These results are inserted in Table 4: comparison between polyphem method speech synthesis and symmetric method speech synthesis:

From Table 4: comparison between polyphem method speech synthesis and symmetric method speech synthesis, we notice that there are varying cases and clear differences in rates, and this is clearly evident with regard to the speech recognition rate for the first and third sentences, which gives a value of 100% using the identical method, while we notice two weak rates in the third test (which is the highest level of smoothed listening). These values are 35.5 and 42.5%, respectively, when using the polyphem method and this represents a clear difference in the results.

We notice that some results are identical for both methods, as we find this in the speech recognition rate for the second sentence at a value of 87.5% and for the fourth and tenth

sentences at a value of 100%. We also notice similarity in the results for the rest of the sentences, except that the results of the polyphem method range between 85.5 and 100%, with 5 iterations at a value of 100%, while the values of the results of the symmetric method are very close to each other, and range between 92 and 100%, with 9 iterations with a value of 100%, which represents twice the number of iterations obtained using the polyphems method.

The important result that can be concluded through this comparison is the overall rate of the regeneration method using the symmetric method, which is estimated at 97.9% with the recorection of two ambiguous sounds in the two sentences (the fourth and seventh) only, that is, 22 listening tests only (20 + 2). In contrast, we find that the results of the poyphem method are taken across three tests, so that the results of modification and correction are taken for twenty sentences in each test, meaning that these modifications are applied to all sentences three times for each sentence, that is, 60 listening tests (20 × 3). This is what provided the results shown in the table; the overall average value is estimated at 90.9%, with an estimated difference of 7% from the overall speech recognition rate

Table 5 Features of some popular TTS systems

Tts software	Reasons to use	Reasons to avoid
NaturalReader	Built-in OCR Choice of interfaces Browser extension Dyslexic-friendly font	Cloud-based Limited free voices compared to paid plans
Balabolka	Excellent file format support Lots of voices to choose from Can create audio files Bookmarking tools	Old-school interface
Panopreter basic	Quick and simple to use Exports in WAV and MP3 formats Good range of input formats	For windows only
World talk	Integrates with microsoft word Customizable voices Speaking dictionary	A little unattractive Microsoft word only
Zabaware text-to-speech reader	Converts text from the clipboard Good file format support	Voices are quite expensive Windows only
Mbrola	Open source artificial voice generation system great degree of In this sense the user can configure various parameters to get precise prosodic control	Control over the synthesized speech at low level
Loquendo TTS	Synthesizes a human-like voice in multiple languages	The community employed it to generate tutorials and parodies
Ivona	TTS system developed by the Amazon company. such as the Kindle electronic reader	It is widely used in Amazon devices only
AT&T	This system developed by the AT&T company It is used in its call centers	It generates speech in eight different languages non-uniform unit selection synthesis with half-phones as units
ATTS using symmetric	Integrates with any text processor For any operating system and Android Free, quick and simple to use Exports in WAV and MP3 formats	Just Arabic language One male voice

for the symmetric method. In addition, the values of the symmetric method are very close to each other, exceeding 92%, which explains the effectiveness, efficiency, distinction, preference, and stabilization of the recognition rate of different sentences of the symmetric method.

3.3 Evaluation of work

The minimum set of sound segments consists of the acoustic units $[CV_h, V_{ho}, V_{hon}, V_{hoe}, V_{hc}, V_{hcn}, V_{hce}]$. These subsyllables included only 124 segments.

We used $3 \times 28 CV_h + 6V_{ho} (3V_{hon} + 3V_{hoe}) + 6V_{hc} (3V_{hcn} + 3V_{hce}) + 28C = 124$ units, in which we have h: half, o: open, c: closed, n: normal and e: emphasized.

By using these units, well-known phonemes in the Arabic language can be generated, especially the units of type [CV] and [CVV]; these two types represent 55% of the units used in the Arabic language [32]; thus, their number is estimated at 168 units that can be generated using the aforementioned phonetic units estimated at only 124 units, which are capable of generating all of the above in high quality, free from sequence distortions.

The number of units used represents only 40% of the number of units used in the automatic synthesis of speech by Diphones, which represents 310 fully used units [32]. The results reached were impressive, as discussed, so that the automatic compound sentences were identified by an estimated rate of 98% using the symmetric method; this is a real breakthrough because this method produced real (normal) units from their real (normal) counterparts.

The pronunciation synergy found in words that contain units of the (AAC) type can be achieved so that we use sounds of the V_{hoe} and V_{hce} types, which contributes to the overall clear highlighting of the effect of emphasisation on the extent of the generated word, which contributes to a generally clear demonstration of the effect of this characteristic on the extent of the generated word using the aforementioned units corresponding to this property.

There are rare cases that occur at a rate of 1% during speech, related to sounds of the [CVCC] type. This indicates that everything involved in the generation of this type of speech has a very small effect on the quality of speech in the Arabic language, given that what we have reached satisfies the purpose and does not limit sound quality ever.

Commercially, there are currently several text-to-speech systems in place, and they largely dominate the global market. To give you an idea of the importance of the technology we propose to ensure the commercialization of this studied system, we present to you Table 5: features of some popular TTS systems, which gives a brief summary of the main features of some popular commercial and experimental text-to-speech (TTS) systems [34–36]:

4 Conclusion

The effectiveness of the phonemes used has a significant impact on the ability to recognize speech synthesis in Arabic. Generative form carrying formant transitions is a very favorable method for the automatic synthesis of speech in the Arabic language. This method is based on the mirror technique, which generates symmetric syllables of the opposite type [CV]. These transitions carry formants to the left and to the right of a vowel; we called this process a left and right linked consonants. This represents a very necessary phase. This phase consists of carrying information on the continuity and intelligibility of the synthesized speech, which is carried out between the consonant and the vowel and is represented acoustically by a variation in the frequencies F1 and F2, called the formants. This information exists in sound units of type [CV] found in 84 possibilities that represent 28 consonants connected with three types of vowels of the Arabic language (Fat'ha [a], Dhamma [u], and Kasra [i]).

In the future, we are interested in a deep learning-based text-to-speech model that synthesizes speech directly from text characters. This is the principle on which the *Tacotron* system is based. We try for working on part of this system by shedding light on deep learning based on the sounds of the Arabic language after the results are obtained. It is based on the use of the analogy method, through which we try to collect speech from the letters of the text without taking into account the characteristics of Arabic texts, which is known as writing. From right to left, we have a method that synthesizes speech on both sides because we care about the letters of the text written in Arabic in both directions.

The results showed that there were good results in terms of the recognition rate of the heard sentences after improving the automatically synthesized speech signal. This suggests that the machine can learn a way to acquire language similar to humans by correcting the combination of the units used through audio corrections. Directly from the translator, this is what the language gains from written texts by extracting their phonetic characteristics corresponding to that position, taking into account the phonetic changes necessary for the written text, which is the work we do when modifying and correcting phonetic units manually, and the WaveNet application helps us in this: which works on waveform classification in integrated speech communication systems while improving accuracy and reducing complexity in human language sounds by projecting the work done by Huynh-The et al. in [27].

We can also mention that the ability to invent a new method for classifying brain tumors [37], enables us to pursue speech pathology, transfer learning-based diagnostic methods and deep learning based on the TTS system.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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